

Thomas Marconi

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EDUCATION

Georgia Institute of Technology

M.S. Computer Science

Atlanta, GA

Expected Dec 2027

- Coursework: Database System Implementation, High-Performance Computer Architecture, Computer Networks

Florida State University

B.S. Computer Science | GPA: 3.7

Tallahassee, FL

May 2023

EXPERIENCE

Tactical Air Support

Software Engineer

Feb 2024 – Present

Aerospace defense contractor supporting U.S. military operations

Jacksonville, FL

- Built a real-time collaborative task platform in **TypeScript/Node.js** with **Socket.io** WebSocket rooms, optimistic UI updates, and server-side reconciliation across a normalized PostgreSQL schema with multi-tenant RBAC, task versioning via full-state JSON snapshots, and a configurable per-user notification rules engine.
- Replaced 3 enterprise vendor systems with purpose-built full-stack tooling (**Next.js**, **FastAPI**, **PostgreSQL**), eliminating **\$850K in annual licensing costs**; architected for minimal-downtime deploys with no dedicated ops team.
- Eliminated static credentials across **8+** dev/prod microservices by implementing runtime secrets injection via **Managed Identity** and **Key Vault**; enforced per-service-principal least-privilege IAM to pass CMMC audit controls.
- Built a RAG pipeline over **5,000+ unstructured PDFs** using **LangGraph** and **Azure Semantic Search**; implemented overlapping semantic chunking and two-stage hybrid retrieval (dense vector + sparse reranking) to improve precision on narrow technical queries.
- Unified Smartsheet and QT9 data sources into a **FastAPI/MongoDB** analytics backend in response to a formal Corrective Action Request; surfaced FPQR trends, CAR classification, and audit forecasting, converting a compliance mandate into a live operational dashboard.

TECHNICAL PROJECTS

Project Desert | *Next.js, Python (FastAPI), PostgreSQL, Docker*

Jan 2025 – Present

- Enforced explicit API contracts via **OpenAPI** schema validation at service boundaries; auto-generated TypeScript clients from **Pydantic models** to surface type violations at compile time and eliminate a class of cross-service integration failures.
- Implemented stateless **JWT authentication** with server-side refresh token rotation across distributed services; engineered to handle concurrent refresh races and session invalidation without shared distributed state.

CityScale | *React, FastAPI, PCA* | **3rd Place — AI for Good Hackathon**

Oct 2025

- Designed and deployed a **PCA-based** housing risk scoring system trained globally across **4,519 census tracts** and 24 curated features, incorporating winsorization, log transforms, and per-capita normalization to produce interpretable 0–100 percentile risk indices across 25 U.S. cities with **66% explained variance** across 6 principal components
- Built a full-stack geospatial affordability visualization platform (**FastAPI + Next.js + Leaflet**) backed by a normalized **SQLite** database of **587,754** data points across **176 indicators**, featuring a real-time user-weighted PCA scoring engine and an **LLM-powered chatbot** for on-demand tract analysis

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL, JavaScript, C++, Java, Bash

Backend: Node.js, FastAPI, WebSockets, REST APIs, PostgreSQL, Docker, Podman

AI & Data: Vector Retrieval (ChromaDB, Azure Semantic Search), LLM Orchestration (LangGraph), Pandas, scikit-learn

Cloud & Infrastructure: Azure (Functions, Managed Identity, Key Vault, Entra ID), CI/CD, Linux

Frontend: React, Next.js, Zustand, TanStack Query, Tailwind CSS